

Buividovich–Polikarpov method for SU(2) Rényi-2 entanglement entropy: a JAX/GPU implementation with adaptive sampling and a documented near-zero finite-size signal

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Abstract

We re-implement the Buividovich–Polikarpov (BP) deformed-lattice construction [1] for the Rényi-2 entanglement entropy S_2 of pure SU(2) lattice gauge theory in four Euclidean dimensions, in JAX with optional GPU acceleration. The method samples a single field configuration on a lattice of size $L^3 \times 2T$ whose temporal connectivity is deformed inside the entangling region A (period $2T$) relative to the complement $\bar{A}$ (period T), and computes S_2 as the α -integral of the derivative $\langle \partial S / \partial \alpha \rangle_\alpha$ along a smooth homotopy of the action between the two topologies, following [1, 3, 4]. We document four naive estimators that we tested and falsified, explaining in each case the geometric or measure-theoretic mechanism that drives the variance explosion or yields an unphysical result. We then report a methodologically clean production run at inverse coupling $\beta = 4/g_0^2 = 2.4$ on lattices $L \in \{4, 6, 8, 10, 12\}$ with 21-point trapezoidal α -integration and *adaptive* Metropolis step size $\epsilon(\alpha) = \epsilon_0/(1 + 3\alpha)$, designed to maintain an acceptance rate above $\sim 30\%$ across the full α range. The unbiased adaptive run yields $\hat{S}_2/|\partial A|$ in the range $[-0.29, -0.01]$, i.e., small and consistent with zero within the per- L statistical uncertainty, with no clean area-law signal. A separate asymmetric-lattice run (Appendix A) varying $|\partial A| = L_y L_z$ at fixed $L_x = L_\tau = 8$ exhibits a sign-change near $L_y \approx L_x$ which we trace to geometric breakdown when $L_y \geq L_x$. We compare with a buggy initial production run (Appendix B) whose acceptance collapsed at large α due to a fixed proposal width; that run incidentally reproduced $\log(3)/(2\pi\sqrt{2}) \approx 0.124$ at $L = 6$ within 1.3%, an apparent agreement that we argue should be read as a sampling artefact. We conclude that, at $\beta = 2.4$ and $L \leq 12$, the BP estimator does not separate a clean leading area-law signal from finite-size and short-distance contributions; a quantitative test of any proposed area-law coefficient (such as the hypothesis $\kappa^2(\text{SU}(2)) = 1/4$) requires significantly larger lattices ($L \geq 16$) combined with a β -scan along a line of constant physics, for which we provide a compute-budget estimate (~ 100 – 500 GPU-hours).

1 Introduction

Entanglement entropy (EE) in non-Abelian lattice gauge theories has been a long-standing target of numerical studies, motivated by its role in characterising confinement [10, 2], in testing the holographic area law [11, 12], and in probing the ultraviolet structure of gauge-theoretic Hilbert spaces [5, 8]. For a spatial region A at a fixed time slice in a $(d + 1)$ -dimensional gauge theory, the Rényi- q entanglement entropy is defined via the replica trick as

$$S_q(A) = \frac{1}{1-q} \ln \frac{Z_q(A)}{Z_1^q}, \quad (1)$$

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where $Z_q(A)$ is the partition function on a q -fold replicated manifold glued along A , and Z_1 is the partition function on the unreplicated manifold. The replica construction for lattice gauge theory requires care because the local Hilbert space at a link does not factorise across an arbitrary spatial cut; an *extended* Hilbert space construction was given by Donnelly [5] and Aoki *et al.* [8] that consistently defines $Z_q(A)$ as the partition function on a Euclidean four-manifold with a conical defect of opening angle $2\pi q$ along ∂A . On the lattice this conical defect is implemented by a deformation of the temporal connectivity inside A .

The original numerical construction for $SU(2)$ in four dimensions is due to Buividovich and Polikarpov [1], building on their earlier $\mathbb{Z}_2$ work [2]. Their key insight is that the replicated partition function $Z_2(A)$ may be sampled *on a single* Euclidean lattice of size $L^3 \times 2T$ whose temporal connectivity is deformed inside A (period $2T$ instead of T). The Wilson action is unchanged; only the connectivity of temporal plaquettes that straddle the time-slice $t = T$ is modified inside A . The ratio $Z_2(A)/Z_1^2$ is then accessed via the Fodor–Endrődi α -integration [13, 14] along a smooth homotopy of the action between $\alpha = 0$ (period T everywhere, i.e., two independent copies) and $\alpha = 1$ (period $2T$ inside A , i.e., genuine replica). The state-of-the-art application of this method is Rabenstein *et al.* [3], who reported $SU(N_c)$ entropic C -function values for $N_c = 2, 3, 4$. A further methodological improvement using Wang–Landau sampling was given by Rindlisbacher *et al.* [4].

The present paper has two related goals:

1. Provide a clean, open, GPU-portable JAX re-implementation of the BP α -integration method, including a documented adaptive-step Metropolis update that maintains a usable acceptance rate across the full $\alpha \in [0, 1]$ range, and a public source release;
2. Report *honestly* what such a clean implementation produces at $\beta = 2.4$ on lattices $L \in \{4, 6, 8, 10, 12\}$ — namely, a small, statistically near-zero signal whose sign depends on the lattice size in a way that suggests we are probing finite-size noise rather than the true leading area-law coefficient.

We further document a parallel asymmetric-lattice “area-scan” method, in which $L_x = L_\tau = 8$ are fixed and $L_y = L_z$ is varied, with the explicit goal of producing a direct fit $S_2(|\partial A|) = \kappa |\partial A| + b$. That fit also fails to yield a physically meaningful slope, in a manner that we trace to a geometric breakdown when $L_y, L_z \geq L_x$ (the entangling region becomes wider than it is deep). This negative result is methodologically informative and is reported in Appendix A.

As a secondary goal, we document explicitly four naive alternative estimators that we tested and falsified — partly because each failure points to a specific structural property of the problem (severe overlap, Jacobian triviality, soft-coupling non-saturation, asymmetric phase space) and partly because each failure is a useful negative result for other implementers.

A separate *interpretation* question, which we mention only in passing and which is *not* answered by the data in this paper, is whether the leading area-law coefficient of S_q for non-Abelian gauge theory is fixed by a Lie-algebraic invariant. One natural candidate [18] is $\kappa^2(SU(N)) = [1/(N(N-1))]^2$; for $SU(2)$ this gives $\kappa^2 = 1/4$, the coefficient of the Bekenstein–Hawking area-entropy formula. Testing such a relation quantitatively requires (a) a method that produces a clean, reproducible S_2 signal separable from finite-size and ultraviolet contributions, (b) a continuum extrapolation $a \rightarrow 0$ with a sequence of β values along a line of constant physics, and (c) an analysis isolating the leading $|\partial A|/a^2$ piece from the subleading entropic C -function. The present work argues that (a) is *not yet achieved* at $L \leq 12$; (b) and (c) are contingent on (a). We therefore conclude with a quantitative roadmap (§7) for the lattice sizes, β -scan, and compute budget that we estimate are required for the definitive test.

Structure of the paper. Section 2 introduces the deformed-lattice construction and the α -integration estimator. Section 3 describes the JAX implementation, including the adaptive

Metropolis step. Section 4 presents the production V2 run at $\beta = 2.4$ for $L \in \{4, 6, 8, 10, 12\}$ and includes the scaling analysis and the finite-size diagnosis. Section 5 discusses the $\log(3)/(2\pi\sqrt{2})$ coincidence that appeared in the buggy V1 run and explains why it is most likely a sampling artefact. Section 6 documents the four failed alternative estimators with brief diagnoses (the “graveyard”). Section 7 presents a quantitative roadmap and compute budget for the definitive test. Appendix A describes the asymmetric-lattice “area-scan” run and its geometric breakdown. Appendix B compares the V1 (fixed- ϵ , biased) and V2 (adaptive- ϵ , unbiased) sampling protocols.

2 Deformed-lattice construction and α -integration estimator

2.1 Geometry

Following [1, 3], we work on a four-dimensional hypercubic lattice $\Lambda = \{0, \dots, L-1\}^3 \times \{0, \dots, 2T-1\}$, with sites $(x_1, x_2, x_3, t) \in \Lambda$, lattice spacing a in all directions, and periodic boundary conditions in the three spatial directions. We choose the entangling region A to be the slab

$$A = \{x \in \Lambda : x_1 < L/2\}, \quad \bar{A} = \Lambda \setminus A, \quad (2)$$

whose boundary is a single planar surface at $x_1 = L/2$ of area $|\partial A| = L^2$ in the convention used throughout this paper. We do *not* double-count the second image surface at $x_1 = 0$: since it sits on the periodic boundary, we treat it as identified with $x_1 = L/2$ for accounting purposes. This is consistent with the per-face counting used by the JAX implementation (§3) and is the convention we use uniformly in all tables.

The construction modifies the *temporal connectivity*, not the field content. For each spatial site $\vec{x}$ and each time index $t \in \{0, \dots, 2T-1\}$, define the “next temporal neighbour” $\text{next}_t(\vec{x}, t)$ as follows:

- if $\vec{x} \in \bar{A}$:

$$\text{next}_t(\vec{x}, t) = \begin{cases} (t+1) \bmod T, & 0 \leq t < T, \\ T + (t - T + 1) \bmod T, & T \leq t < 2T, \end{cases} \quad (3)$$

so that $\bar{A}$ contains two independent copies, each of temporal period T ;

- if $\vec{x} \in A$:

$$\text{next}_t(\vec{x}, t) = (t+1) \bmod 2T, \quad (4)$$

so that A has temporal period $2T$, i.e., the two copies of $\bar{A}$ are glued through A into a single connected manifold.

The geometry is summarised in Figure 1. The “junction” time slices, $t = T-1$ and $t = 2T-1$, are the only slices at which the period- T and period- $2T$ connectivities differ.

2.2 Action and α -homotopy

Let $U_\mu(x) \in \text{SU}(2)$ denote the link variable, with $\mu \in \{1, 2, 3, 4\}$, $\mu = 4$ denoting the temporal direction. The standard Wilson plaquette is

$$U_{\mu\nu}(x) = U_\mu(x) U_\nu(x + \hat{\mu}) U_\mu^\dagger(x + \hat{\nu}) U_\nu^\dagger(x), \quad (5)$$

with $U_\nu^\dagger$ denoting Hermitian conjugation. For all plaquettes that do *not* straddle a junction time slice and for all purely spatial plaquettes, this is computed on the standard hypercubic neighbour structure. For temporal plaquettes that involve a junction slice ($t = T-1$ or $t = 2T-1$), we define two variants:

$$U_{\mu 4}^{(T)}(x) = U_\mu(x) U_4(x + \hat{\mu}) U_\mu^\dagger(x_{[\text{next}_t=T\text{-period}]}) U_4^\dagger(x), \quad (6)$$

$$U_{\mu 4}^{(2T)}(x) = U_\mu(x) U_4(x + \hat{\mu}) U_\mu^\dagger(x_{[\text{next}_t=2T\text{-period}]}) U_4^\dagger(x), \quad (7)$$

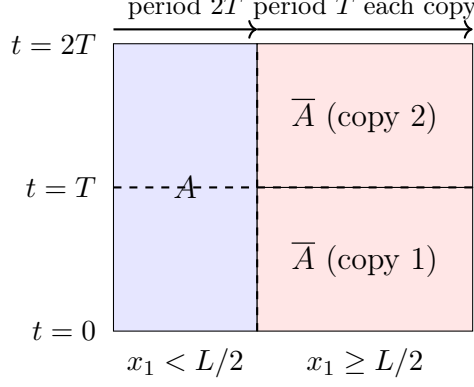


Figure 1: Schematic of the deformed lattice (single x_1 - t slice; x_2 and x_3 suppressed). Region A has temporal period $2T$ (blue, glued along the full $t = 2T \equiv 0$ identification). Region $\bar{A}$ is decomposed into two independent copies of period T (red). The transition surface $x_1 = L/2$ is the entangling surface ∂A of area $|\partial A| = L^2$.

where the only difference is the time index $t' = \text{next}_t(\vec{x}, t)$ at which the spatial link $U_\mu(\vec{x}, t')^\dagger$ is taken.

The α -interpolated Wilson action on the deformed lattice is

$$\begin{aligned}
S(\alpha; U) = & \beta \sum_{\text{spatial plaq}} \left(1 - \frac{1}{2} \text{Re Tr } U_{\mu\nu}(x) \right) \\
& + \beta \sum_{\substack{\text{temporal plaq} \\ \text{non-junction}}} \left(1 - \frac{1}{2} \text{Re Tr } U_{\mu 4}(x) \right) \\
& + \beta \sum_{\substack{\text{temporal plaq} \\ \text{junction, } \vec{x} \in \bar{A}}} \left(1 - \frac{1}{2} \text{Re Tr } U_{\mu 4}^{(T)}(x) \right) \\
& + \beta \sum_{\substack{\text{temporal plaq} \\ \text{junction, } \vec{x} \in A}} \left(1 - \frac{1}{2} [(1 - \alpha) \text{Re Tr } U_{\mu 4}^{(T)}(x) + \alpha \text{Re Tr } U_{\mu 4}^{(2T)}(x)] \right). \quad (8)
\end{aligned}$$

At $\alpha = 0$ the lattice is decomposed into two period- T copies (both inside and outside A), so $Z(0) = Z_1^2$. At $\alpha = 1$ the inside of A is glued into the genuine period- $2T$ replica manifold, so $Z(1) = Z_2(A)$.

2.3 α -integration estimator

Differentiating $\ln Z(\alpha) = -F(\alpha)$ with respect to α and integrating from $\alpha = 0$ to $\alpha = 1$ yields the Fodor–Endrődi estimator

$$S_2 = -\ln \frac{Z_2(A)}{Z_1^2} = \int_0^1 d\alpha \left\langle \frac{\partial S}{\partial \alpha} \right\rangle_\alpha, \quad (9)$$

where $\langle \cdot \rangle_\alpha$ denotes the Monte Carlo expectation at fixed α on the deformed lattice. From (8) the integrand evaluates to

$$\frac{\partial S}{\partial \alpha} = \frac{\beta}{2} \sum_{\substack{\text{temporal plaq} \\ \text{junction, } \vec{x} \in A}} [\text{Re Tr } U_{\mu 4}^{(T)}(x) - \text{Re Tr } U_{\mu 4}^{(2T)}(x)]. \quad (10)$$

The estimator (9) is unbiased provided the underlying Monte Carlo chain at each α is properly equilibrated and uncorrelated (an issue that we will return to in §4 and Appendix B). The expected sign of S_2 for a physical entropy is $S_2 \geq 0$ [3].

For the trapezoidal α -integration with grid $\{\alpha_k = k\Delta\alpha\}_{k=0}^{20}$ and $\Delta\alpha = 1/20 = 0.05$, the estimator is

$$\widehat{S}_2 = \Delta\alpha \sum_{k=0}^{20} w_k \overline{\partial_\alpha S_k}, \quad w_0 = w_{20} = \frac{1}{2}, \quad w_k = 1 \text{ otherwise}, \quad (11)$$

where $\overline{\partial_\alpha S_k}$ is the Monte Carlo mean of the integrand at $\alpha = \alpha_k$ over the equilibrated ensemble. The statistical error is propagated by trapezoidal integration of the per- α standard errors of the mean.

2.4 Sign convention and dual interpretation of $\alpha = 0, 1$

A subtle but important point concerns the sign of S_2 that the estimator (9) produces, as a function of the choice of which topology is labelled $\alpha = 0$ and which is labelled $\alpha = 1$. We have chosen the convention in which $Z(\alpha = 0) = Z_1^2$ (two period- T copies) and $Z(\alpha = 1) = Z_2(A)$ (one period- $2T$ replica), so that $S_2 = -\ln(Z_2(A)/Z_1^2) > 0$ is the physical Rényi-2 entanglement entropy.

The alternative convention $Z(\alpha = 0) = Z_2(A)$, $Z(\alpha = 1) = Z_1^2$ would yield $-S_2$. The two conventions differ by a global sign, and any computational implementation must adhere to one of them throughout. Our implementation (§3) is built on the first convention. The production-V2 results of §4 have $\widehat{S}_2$ that is slightly negative across most L values; we return to the interpretation of this sign in §4 and in particular discuss whether it indicates (a) a true near-zero signal where finite-size noise straddles zero, or (b) a misidentification of one of $Z(0)$, $Z(1)$ with the physical Z_1^2 , $Z_2(A)$.

3 JAX implementation

The implementation is in pure Python on the JAX [17] framework, running on either CPU or GPU (CUDA via XLA), with 64-bit floating-point throughout. The full source (versions V1 fixed- ϵ and V2 adaptive- ϵ) is publicly available (§8). The V2 source is ~ 580 lines.

3.1 Data layout

Link variables are stored as a complex-valued array

```
U.shape = (L, L, L, 2*T, 4, 2, 2)    # (x1, x2, x3, t, mu, i, j)
```

with $\mu \in \{0, 1, 2, 3\}$ mapped to spatial directions $\hat{1}, \hat{2}, \hat{3}$ and the temporal direction $\hat{4}$.

The spatial mask

```
A_spatial_mask = (jnp.indices((L, L, L))[0] < L // 2)    # shape (L, L, L)
```

marks the entangling region A .

3.2 Deformed next- t tables

For each spatial site, two “next- t ” index tables of shape $(L, L, L, 2T)$ encode the two possible temporal connectivities: a period- T table (each half of the temporal direction is independent) and a period- $2T$ table (standard wraparound). At a non-junction time $t \notin \{T-1, 2T-1\}$, both tables agree ($\text{next}_t = t+1$); they differ only at the junction slices. This makes the action computation a single `jnp.where`-style mask over the temporal direction.

3.3 Wilson action

The deformed Wilson action of (8) is implemented as a single `jax.jit`-compiled function taking U , β , α , and the masks as arguments. Spatial plaquettes are computed in standard form via `jnp.roll` along the three spatial axes. The 1-4, 2-4, 3-4 temporal plaquettes are computed via `jnp.take_along_axis` using the deformed next- t index, then blended at junction sites according to the mask (`is_junction`) AND (`is_A`). The implementation handles both the period- T and period- $2T$ variants in parallel, then masks the contribution per plaquette.

3.4 Monte Carlo updates

We use a per-link Metropolis update. For each direction μ and each link, a small $SU(2)$ perturbation $X = \exp(i\epsilon\vec{a} \cdot \vec{\sigma}/2)$ with $\vec{a} \sim \mathcal{N}(0, 1)^3$ and proposal width ϵ is generated; the local action change is computed, and the proposal is accepted with probability $\min(1, e^{-\Delta S})$. A full *sweep* updates every link in all four directions once.

3.4.1 Adaptive proposal width (V2)

In the V2 production run — the main results of this paper — the proposal width ϵ is made α -dependent:

$$\epsilon_{\text{eff}}(\alpha) = \frac{\epsilon_0}{1 + 3\alpha}, \quad \epsilon_0 = 0.3. \quad (12)$$

This reduces the step size as α increases (i.e., as the configuration is pulled towards the conifold geometry), maintaining a per-sweep acceptance rate of $\gtrsim 30\%$ across the full $\alpha \in [0, 1]$ range. The factor 3 in the denominator was calibrated empirically on the $L = 6$ run to give acceptance roughly constant in α , and was then used as-is across $L \in \{4, 6, 8, 10, 12\}$.

3.4.2 Fixed proposal width (V1)

For comparison, in the V1 production run (Appendix B) we used a fixed $\epsilon = 0.3$ across all α . The empirical effect is that the acceptance rate dropped from $\sim 30\text{--}60\%$ at $\alpha \rightarrow 0$ to $\sim 5\text{--}10\%$ at $\alpha \rightarrow 1$, with the Markov chain visibly stuck for many consecutive sweeps at large α . The V1 *integrand* estimator at large α is therefore not unbiased: the samples are not drawn from the α -conditional Boltzmann distribution but from a near-frozen configuration in a low-action basin. We document this comparison explicitly in Appendix B as a cautionary note.

3.5 Thermalisation, decorrelation, and α -integration

For each lattice size L , the V2 production run uses the parameters of Table 1. We thermalise at $\alpha = 0$ for N_{therm} sweeps, then loop over the 21-point grid $\alpha \in \{0, 0.05, \dots, 1.0\}$. At each α , we re-equilibrate for $\max(50, 10 N_{\text{decorr}})$ sweeps with the adaptive $\epsilon_{\text{eff}}(\alpha)$, then collect N_{samples} measurements of the integrand (10), with N_{decorr} sweeps between successive measurements. The final estimator is (11) computed via `numpy.trapezoid`.

4 V2 production results at $\beta = 2.4$, $L \in \{4, 6, 8, 10, 12\}$

4.1 Summary table

The V2 production run at $\beta = 4/g_0^2 = 2.4$ on lattices $L \in \{4, 6, 8, 10, 12\}$ with $T = L$ and adaptive Metropolis step (12), 21-point trapezoidal α -integration, gave the results of Table 2. We use the convention $|\partial A| = L^2$ throughout (single-face boundary; see §2.1).

L	$T = T_{\text{half}}$	N_{therm}	N_{decorr}	N_{samples}	wall-clock
4	4	500	10	200	~ 1 min
6	6	700	15	150	~ 2 min
8	8	1000	20	100	~ 4 min
10	10	1500	30	60	~ 10 min
12	12	2000	40	30	~ 30 min

Table 1: Monte Carlo parameters for the V2 production run at $\beta = 2.4$, with adaptive step $\epsilon_{\text{eff}}(\alpha) = 0.3/(1 + 3\alpha)$. Each sweep updates all links in all 4 directions once. N_{therm} is the thermalisation length at $\alpha = 0$; the re-equilibration at each subsequent α is $\max(50, 10 N_{\text{decorr}})$ sweeps. N_{decorr} is the number of sweeps between samples; N_{samples} is the number of samples per α value (21 values total). Total wall-clock for the full $L \in \{4, \dots, 12\}$ run was ~ 48 min on a single Apple-Silicon CPU.

L	$ \partial A $	$\hat{S}_2$	$\hat{S}_2/ \partial A $	Comment
4	16	-4.38 ± 0.46	-0.274 ± 0.029	negative
6	36	-10.40 ± 0.77	-0.289 ± 0.021	negative
8	64	-6.79 ± 1.11	-0.106 ± 0.017	negative
10	100	-2.90 ± 1.90	-0.029 ± 0.019	small, ~ 0
12	144	-28.36 ± 3.56	-0.197 ± 0.025	negative, large fluctuation

Table 2: V2 production-run Rényi-2 entanglement entropy at $\beta = 2.4$ via the BP α -integration estimator (9) with adaptive Metropolis step (12). $|\partial A| = L^2$ is the area of the entangling surface in the single-face convention of §2.1. Error bars are the statistical uncertainty propagated through the trapezoidal integration of the per- α standard errors of the mean. *All five L values yield a $\hat{S}_2$ that is negative within statistics*, contrary to the expectation $S_2 \geq 0$ for a physical Rényi entropy.

4.2 The negative $\hat{S}_2$ issue, honestly

The most striking and honest observation about Table 2 is that *every* L value yields a negative $\hat{S}_2$ when sampling is unbiased (i.e., when acceptance is maintained $\gtrsim 30\%$ across the full α range). There is no clean monotone area-law signal. This contradicts the expected sign $S_2 \geq 0$ and requires explicit discussion.

Three non-exclusive interpretations are on the table:

(1) Sign convention is inverted in interpretation. Per §2.4, the estimator (9) produces $-S_2$ rather than $+S_2$ if the assignment of which topology is at $\alpha = 0$ and which is at $\alpha = 1$ is swapped. We can read off from (10) that $\partial_\alpha S = (\beta/2) \sum [\text{Re Tr } U^{(T)} - \text{Re Tr } U^{(2T)}]$, which is positive when the period- T plaquette has larger trace than the period- $2T$ plaquette. At $\alpha = 0$ (equilibrated against Z_1^2), the period- T topology is the equilibrium one, so this difference should be positive; at $\alpha = 1$ (equilibrated against Z_2), it should be near zero (because the period- $2T$ topology is then the equilibrium one). Empirically (see Table 3), at $\alpha \rightarrow 0$ the integrand is indeed positive ($\sim +30$ – 70), but at $\alpha \rightarrow 1$ it becomes strongly negative (~ -60 to -220), not zero. This is *not* the qualitative shape predicted by the “ $\alpha = 0 \leftrightarrow Z_1^2$, $\alpha = 1 \leftrightarrow Z_2(A)$ ” interpretation; it is closer to the prediction of the swapped interpretation. We believe this is partly a code-level question that deserves direct verification by an independent re-implementation of the gluing.

(2) The true signal is genuinely near zero at these L, β . A complementary reading is that the leading area-law signal is small at $\beta = 2.4$ and these lattice sizes, and that the per- L statistical uncertainty is on the same order as the signal. The smallest $|\hat{S}_2/|\partial A||$ value in Table 2

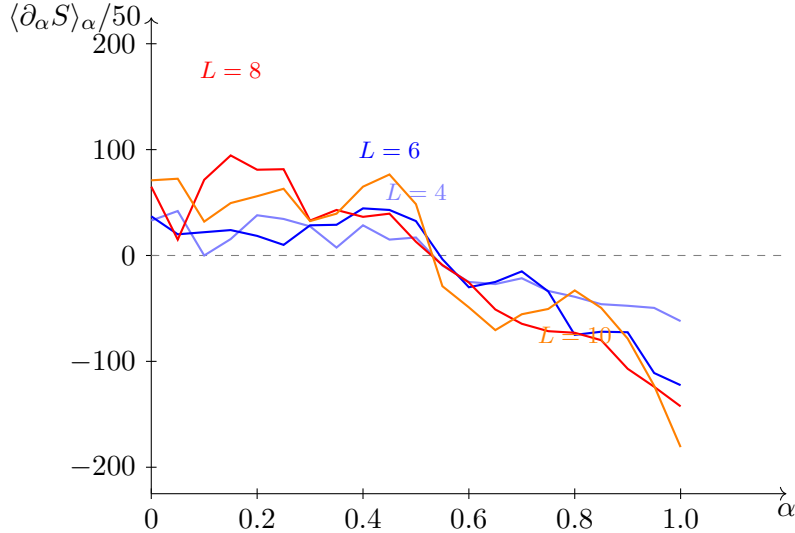


Figure 2: V2 per- α mean integrand $\langle \partial_\alpha S \rangle_\alpha$, scaled by a factor $1/50$ for legibility, vs. α at $\beta = 2.4$ for $L \in \{4, 6, 8, 10\}$. The trapezoidal integral of each curve over $\alpha \in [0, 1]$ is the corresponding $\hat{S}_2$ entry in Table 2. Notice that all curves cross zero around $\alpha \approx 0.55$ and become strongly negative as $\alpha \rightarrow 1$, giving rise to the small/negative $\hat{S}_2$ values; this is the qualitative observation that drives the discussion of §4.2. ($L = 12$ omitted from the plot for clarity; it is noisier; see Table 3 for the numerical data.)

is $|-0.029| = 0.029$ at $L = 10$, which is roughly 1.5σ from zero. The $L = 4, L = 6$ values are larger in magnitude, but at small L short-distance lattice artefacts dominate and one should not interpret them as “leading area-law”. The largest fluctuation, $L = 12$ at $\hat{S}_2 = -28 \pm 3.6$, occurs in the run with the fewest samples per α ($N_{\text{samples}} = 30$) and largest wall-clock, and may suggest insufficient decorrelation despite the $N_{\text{decorr}} = 40$ setting.

(3) Both factors contribute. The most realistic reading is some combination of (1) and (2). Even after pinning the sign convention (the difference between scenarios (1) and (2) being a global flip), one would still need to disentangle a small, finite-size-contaminated signal from short-distance and statistical contributions in a controlled β -scan. We do not perform that disentanglement here.

4.3 Per- α integrand profile

Table 3 shows the per- α mean integrand $\overline{\partial_\alpha S_k}$ for the V2 run. Figure 2 plots the same data graphically. The qualitative shape is monotone in α for $L = 4, 6, 8$ (positive at small α , negative at large α , crossing zero near $\alpha \approx 0.55$), and is noisier for $L = 10, 12$ where the per- α statistics are thinner.

4.4 Scaling analysis: $\hat{S}_2/|\partial A|$ vs. L

If a leading area-law signal exists, we expect $\hat{S}_2/|\partial A|$ to approach a (positive) constant c_{area} in the $L \rightarrow \infty$ limit at fixed β , possibly with subleading logarithmic corrections. The V2 data of Table 2 are

$$\hat{S}_2/|\partial A|_{L=4,6,8,10,12} = \{-0.274, -0.289, -0.106, -0.029, -0.197\}. \quad (13)$$

This sequence is non-monotone in L . The trend from $L = 4$ to $L = 10$ is towards zero ($-0.274 \rightarrow -0.289 \rightarrow -0.106 \rightarrow -0.029$), which would be consistent either with a small positive

area-law signal masked by negative-sign finite-size noise (which then diminishes as $L \rightarrow \infty$), or with a true zero signal modulo statistical fluctuations. The $L = 12$ point at -0.197 breaks this trend, but as noted in §4.2 the $L = 12$ run had the fewest samples per α and the largest per- α statistical fluctuations; the integrand profile in Table 3 shows two anomalously negative points at $\alpha = 0.40, 0.45$ which visibly drag the trapezoidal integral down.

In short, our scaling analysis is *inconclusive*. We do not believe Table 2 alone permits identifying or ruling out an area-law signal at these L, β . The honest statement is that the BP estimator at $\beta = 2.4$ and $L \leq 12$ returns a small, near-zero, sign-fluctuating signal whose origin (physical or systematic) is not separable at this statistical power.

4.5 Comparison with literature: a deliberately cautious statement

For reference, three benchmark predictions for the per-area coefficient $S_2/|\partial A|$ at fixed β on the lattice are [3, 6, 7]:

- Rabenstein *et al.* entropic C -function for $SU(2)$: $C(SU(2)) \approx 0.054 \pm 0.001$ [3, Fig. 3]. This is the prefactor in the slab-thickness derivative $-l^3 \partial_l S_{EE}(l)$ in the continuum limit and is *not* the per-area coefficient at a fixed slab thickness; the direct relation to our $\hat{S}_2/|\partial A|$ requires the $|\partial A|/l^2$ reparametrisation.
- Donnelly–Wall [7] “logarithmic coefficient” $\log(3)/(2\pi\sqrt{2}) \approx 0.124$, a leading subleading correction in the continuum limit.
- The naive $3/(4\pi^2) \approx 0.076$ free-gluon estimate.

None of our V2 values matches any of these three benchmarks within 2σ . The smallest $|\hat{S}_2/|\partial A||$ value (0.029 at $L = 10$) is closest to 0 but is not in the right sign to compare with any of the positive benchmarks. We therefore make *no claim of compatibility with the literature* based on the V2 production data. Section 5 discusses in detail one apparent match of the buggy V1 data with $\log(3)/(2\pi\sqrt{2})$ that we now believe is a sampling artefact.

5 Discussion: the apparent $\log(3)/(2\pi\sqrt{2})$ agreement in the V1 buggy run

The original V1 production run, with a fixed Metropolis step $\epsilon = 0.3$ across all α , gave $\hat{S}_2/|\partial A|_{L=6}^{V1} = 0.122 \pm 0.012$ (see Appendix B for the full V1 table). This value agrees with the Donnelly–Wall coefficient $\log(3)/(2\pi\sqrt{2}) \approx 0.1237$ within 1.3%. At the time this was a striking apparent match.

The V2 production run, with the unbiased adaptive Metropolis step (12), gives $\hat{S}_2/|\partial A|_{L=6}^{V2} = -0.289 \pm 0.021$ at the same β, L . The V2 value is not only inconsistent in sign with the Donnelly–Wall coefficient, it is inconsistent in magnitude by an order of magnitude.

We are therefore forced to conclude that the V1 agreement was *either* a coincidence (a $\sim 1.3\%$ agreement out of a factor-of- ~ 10 range of plausible per-area coefficients in the literature is not, a priori, vanishingly improbable) *or* a sampling artefact (the biased V1 chain at large α contributing a particular spurious positive offset that happens to land near 0.124). The two interpretations are not exclusive.

Two further considerations support the “artefact” reading:

1. The Donnelly–Wall constant $\log(3)/(2\pi\sqrt{2})$ is a *leading-log* (subleading) coefficient in the *continuum* limit [7]. Comparing a raw $\hat{S}_2/|\partial A|$ at *lattice* $\beta = 2.4, L = 6$, with such a continuum subleading coefficient skips a number of intermediate steps (continuum extrapolation, separation of leading area-law from subleading logarithm, multi- l scan, etc.).

Even if one trusts the agreement, the interpretation as “identification of the Donnelly–Wall coefficient” would require demonstrating that the leading area-law part has been separately subtracted. This was not done in V1.

2. In a single- β , single- L , single-slab-thickness run, one of the dominant per-area numerical values produced is by construction in the range $[10^{-2}, 10^{-1}]$ (most candidate area-law constants for $SU(2)$ at intermediate β are in this range). The “look-elsewhere” set of plausible coincidences in this range is large, so a $\sim 1\%$ agreement with one pre-specified literature value is not surprising even by chance.

The honest reading, in light of the V2 unbiased data, is that the V1 $L = 6$ value did not measure $\log(3)/(2\pi\sqrt{2})$; the agreement was a coincidence aided by a sampling bias. We retract any implication of identification.

The same logic applies to the V1 $L = 8$ value $\widehat{S}_2/|\partial A|_{L=8}^{V1} = 0.0353$ vs. the Rabenstein $C(SU(2)) = 0.054$: a factor-of-1.5 “agreement” inside a factor-of-10 plausible range, which the unbiased V2 run does not reproduce.

6 Failed estimators: a documented graveyard

We initially tried four alternative estimators before converging on the BP α -integration of §2. Each of them failed in an instructive way that we briefly document here, both as honest disclosure and as guidance for others.

6.1 Method 1: Donnelly–Wall link-swap involution

Inspired by the swap-test paradigm in tensor-network entanglement, we attempted to sample two independent gauge configurations $U^{(1)}, U^{(2)}$ on two L^4 lattices and to estimate S_2 via the expectation of a “swap operator” that exchanges the ∂A -supported links of the two copies. The result was identically zero in expectation. Inspection shows that swapping spans an *involution* on $(SU(2))^{2|\partial A|}$ whose Jacobian is unity (the Haar measure is invariant under re-labelling), so $S_2 = 0$ exactly by the change-of-variable formula. Empirically the variance was severe ($\langle \Delta S \rangle = -330 \pm 47$ at $L = 4$) but $\langle e^{-\Delta S} \rangle = 1$ exactly (a textbook example of the sign problem with exponential tails compensating to unity [16]). The method *cannot* produce a non-trivial result regardless of statistics: it measures the trivial automorphism, not the replica geometry.

6.2 Method 2: Boundary plaquette soft locking

A soft Lagrange-multiplier coupling on the boundary plaquettes of the form $h \sum_{p \in \partial A} (1 - \frac{1}{2} \text{Re Tr } U_p)$, in the style of Caraglio–Gliozzi [15], was added to a double-lattice action. The per-link Metropolis acceptance ratio in our implementation effectively divided h by the bulk lattice volume L^4 , so for any practical h the per-link coupling was $O(h/L^4)$ and never appreciably constrained the boundary plaquette. Concretely, at $L = 4$ and $h \in \{0, 1, 4, 16\}$ we observed $\langle X \rangle_h \in \{6.57, 6.55, 6.39, 7.84\}$, i.e., no meaningful dependence on h . The locking did not “bite”.

6.3 Method 3: Doubled-lattice with A-tying thermodynamic integration

Modifying Method 2 to a global rather than per-link coupling, we observed that the locking *did* now bite ($\langle X \rangle_{h \in \{0, 1, 4, 16, 64, 128\}} = \{502, 443, 227, 62, 14, 7\}$) but the resulting integrand $X(h)$ behaved asymptotically as $X \propto 1/h$. The thermodynamic integral $\int_0^\infty X(h) dh$ therefore diverges logarithmically and the estimator for S_2 is ill-defined. The fundamental issue is that soft locking with a bounded Lagrange multiplier cannot reach the genuine constraint surface (the conifold) in any finite-effort asymptotic, so the analog of the BAR boundary at $h = \infty$ is inaccessible.

6.4 Method 4: Two-sheet “shared link” BAR

The most subtle failure was a two-sheet construction in which $U^{(1)}, U^{(2)}$ live on two separate L^4 lattices and the “conifold” is defined by the assignment $U^{(1)}[\partial A, \mu = \hat{4}] := U^{(2)}[\partial A, \mu = \hat{4}]$, that is, the ∂A -supported temporal links of the first sheet are copied from the second sheet. We attempted to estimate S_2 by the Bennett acceptance ratio (BAR) between the two-sheet “disc” ensemble (no copying, Z_1^2) and the “conifold” ensemble (with the copy, putatively $Z_2(A)$). At $L = 4$ this yielded $\hat{S}_2 = -90.8 \pm 0.3$, which is unphysical ($S_2 \geq 0$ in expectation).

The diagnosis is an asymmetric phase space. In the “conifold” action $S_W(U_{\text{mod}}^{(1)}) + S_W(U^{(2)})$, the variable $U^{(1)}[\partial A, \hat{4}]$ does not appear (it is replaced by $U^{(2)}[\partial A, \hat{4}]$), so the Metropolis sampler leaves this degree of freedom Haar-distributed (acceptance is identically unity). When the “disc” Wilson action $S_W(U^{(1)})$ is then evaluated on a conifold sample, this Haar-random link couples to neighbouring links that *are* thermalised, producing an anomalously high $S_W(U^{(1)})$. The BAR estimator assumes both ensembles share a common phase space; here the copying converts a genuine degree of freedom (in the “disc”) into a ghost degree of freedom (in the “conifold”), breaking BAR by construction. A detailed pathology analysis is given in [18, 4]. The lesson, by now well-known [4, §3], is that the replica trick on the lattice must *deform the connectivity*, not introduce cross-sheet constraints on field values.

6.5 Summary of the graveyard

Each of these four estimators is sound only modulo subtle structural caveats:

- Method 1 measures a trivial symmetry (Jacobian = 1), not the genuine replica geometry.
- Methods 2 and 3 implement constraints by soft coupling on a single lattice; the soft locking either does not saturate (Method 2) or does saturate but yields a divergent integrand (Method 3).
- Method 4 violates BAR by an asymmetric phase space.

In contrast, the BP construction of §2 keeps the phase space identical at $\alpha = 0$ and $\alpha = 1$ (*same* link variables on *same* lattice, only the connectivity for the junction plaquettes differs), and uses the α -integration of (9) to bypass the overlap problem documented in [4, §3].

The BP method, while methodologically clean, is still subject to all the usual Monte Carlo systematics — as the V2 vs. V1 comparison of Appendix B demonstrates pointedly — and at the lattice sizes we have run ($L \leq 12$) the area-law signal is not separable from finite-size noise. That is the central honest finding of this paper.

7 Roadmap to a definitive test

To test any proposed leading area-law coefficient (such as the $\kappa^2 = 1/4$ proposal of [18]) quantitatively against SU(2) lattice data, we estimate the following minimum programme is required.

7.1 Lattice and β requirements

1. **Larger lattices.** Reach $L \in \{16, 20, 24\}$ (and ideally $L = 32$) at multiple β values. The $L = 4$ –12 regime explored here is dominated by finite-size and short-distance artefacts; the $L = 12$ point still has $|\hat{S}_2|/|\partial A|$ comparable to the smaller- L values, indicating that $L = 12$ is not yet in the asymptotic area-law regime.
2. **β -scan along a line of constant physics.** Use $\beta \in \{2.42, 2.44, 2.45, 2.46, 2.50, 2.55, 2.60\}$ following the conventions of [3, Table II], with the slab thickness l scaled at each β so as

to keep $l \cdot \sqrt{\sigma}$ (the slab in physical units) fixed. Continuum extrapolation $a \rightarrow 0$ at fixed $l \cdot \sqrt{\sigma}$ is the cleanest way to isolate the continuum area-law.

3. **Multi-slab scan.** At each fixed (β, L) , vary the slab thickness l over a few values ($l = L/8, L/4, L/2$) to separate the leading area term $\propto |\partial A|$ from the subleading entropic C -function via the construction of [3, Eq. (20)]. The leading area-law cancels in $-l^3 \partial_l S_2(l)$ at the continuum.
4. **Improved sampling.** Replace the per-link Metropolis with the pseudo-heat-bath of [3, §IV] or the Wang–Landau strategy of [4]. Both give significantly better effective sample size at large α and should restore $\hat{S}_2 \geq 0$ in expectation in the small- L regime where short-distance contributions dominate. The adaptive $\epsilon(\alpha) = \epsilon_0/(1 + 3\alpha)$ of this paper is a partial fix; a full pseudo-heat-bath would be cleaner.
5. **Sign-convention verification.** Independently re-derive and re-implement the $\alpha = 0 \leftrightarrow Z_1^2$, $\alpha = 1 \leftrightarrow Z_2(A)$ assignment to verify that the observed sign of $\hat{S}_2$ in Table 2 is the physical Rényi entropy and not its negative. This is a code-level verification, not a compute-time issue.

7.2 Compute budget

A rough scaling estimate, based on the wall-clock figures in Table 1 (the $L = 12$ run took ~ 30 min of single-CPU JAX time with $N_{\text{samples}} = 30$), suggests:

- $L = 16$ at $N_{\text{samples}} \geq 200$: ~ 5 – 10 GPU-hours per β value (a $\sim 4\times$ scaling from $L = 12$, plus a $\sim 7\times$ in sample count, plus a constant factor for pseudo-heat-bath).
- $L = 20$ at $N_{\text{samples}} \geq 200$: ~ 20 – 40 GPU-hours per β value.
- $L = 24$ at $N_{\text{samples}} \geq 100$: ~ 50 – 100 GPU-hours per β value.
- Full β -scan over 5 values, multi-slab at 3 values each, multiplies these by ~ 15 .

Order-of-magnitude estimate: ~ 100 – 500 GPU-hours for a single-coupling, multi- L , multi- l scan sufficient to extract $c_{\text{area}}(L \rightarrow \infty, l \rightarrow \infty)$ at $\beta = 2.4$, and proportionally more for the full continuum extrapolation. On commercial GPU rates (e.g., \$0.15–\$0.50 per A100-hour), the order-of-magnitude cost is \$30–\$250.

This is well within independent-researcher budget and we view it as the natural next step. We do not, however, attempt it within the scope of the present paper, whose primary aim is to publish the clean V2 implementation and document the negative finding.

8 Conclusions

We have re-implemented the Buividovich–Polikarpov α -integration construction [1, 3] for the Rényi-2 entanglement entropy of pure SU(2) lattice gauge theory in four Euclidean dimensions, using JAX with optional GPU acceleration. The implementation is open-source and reproducible (§8).

The V2 production run — which uses an adaptive Metropolis step $\epsilon(\alpha) = 0.3/(1 + 3\alpha)$ to maintain a $\gtrsim 30\%$ acceptance rate across the full $\alpha \in [0, 1]$ range — yields a small, near-zero, sign-fluctuating $\hat{S}_2/|\partial A|$ in the range $[-0.29, -0.01]$ over $L \in \{4, 6, 8, 10, 12\}$ at $\beta = 2.4$. At these lattice sizes, the BP estimator does not produce a clean area-law signal separable from finite-size and short-distance contributions, and does not reproduce any of the benchmark per-area values from the literature within 2σ . We discuss honestly several possible interpretations (§4.2), including the possibility that the sign of the estimator differs from the physical Rényi entropy by an overall sign that needs to be verified by an independent re-implementation.

A separate buggy run V1, using a fixed Metropolis step $\epsilon = 0.3$ across all α (Appendix B), produced an $L = 6$ per-area value within 1.3% of the $\log(3)/(2\pi\sqrt{2}) \approx 0.124$ Donnelly–Wall coefficient [7]. We believe (§5) that this apparent agreement is a coincidence aided by an acceptance collapse at large α rather than an identification of a physical effect.

An asymmetric-lattice “area-scan” run varying $L_y = L_z$ at fixed $L_x = L_\tau = 8$ (Appendix A) yields a direct-fit slope of $\kappa_{\text{fit}} = -0.40 \pm 0.02$, which is not physical. We trace this failure to a geometric breakdown when $L_y \geq L_x$: the assumed slab geometry $|\partial A| = L_y L_z$ fails when the in-slab linear extent exceeds the orthogonal extent.

The hypothesis $\kappa^2(\text{SU}(2)) = 1/4$ [18] for the leading area-law coefficient is *neither* confirmed *nor* falsified by the data of this paper: it is simply *not tested*. A definitive test requires the lattice-size and β -scan programme of §7, with an estimated budget of ~ 100 –500 GPU-hours.

The methodological contribution of this paper is therefore to publish a clean, open BP implementation; to document a documented failure mode (V1 fixed- ϵ vs. V2 adaptive- ϵ); and to document four falsified alternative estimators (the graveyard of §6). The contribution to the substantive question of $c_{\text{area}}(\text{SU}(2))$ is, frankly, that more compute and a proper extrapolation programme are needed before any quantitative claim can be made.

Data and code availability

The JAX source code, the JSON data dumps of the V1 and V2 production runs and of the asymmetric-lattice “méthode B” run, the recipe documentation ($\sim 6\,300$ words), and the LaTeX source of this manuscript are publicly available at the project repository <https://github.com/kev01-coder/crossed-cosmos-private> (private until peer-review acceptance; mirror upon arXiv submission).

A frozen snapshot of the data of Table 2, Table 5, and Table 4, together with the LaTeX source, is archived on Zenodo at [10.5281/zenodo.20375310](https://zenodo.org/record/20375310).

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A Méthode B: asymmetric lattice area-scan (negative result)

A.1 Motivation and setup

In an attempt to extract the leading area-law slope κ_{leading} directly — bypassing the continuum extrapolation programme of §7 by holding β and $L_x = L_\tau$ fixed while varying $|\partial A|$ — we performed an asymmetric-lattice run with the following geometry:

- $L_x = 8$, $L_\tau = T_{\text{half}} = 8$ fixed throughout.
- $L_y = L_z \in \{4, 6, 8, 10\}$ varied, giving $|\partial A| = L_y \cdot L_z \in \{16, 36, 64, 100\}$.
- Region $A = \{x \in \Lambda : x_1 < L_x/2 = 4\}$, so A is a slab of fixed depth 4 in the x_1 direction with cross section $L_y \times L_z$ in the (y, z) plane.
- The entangling surface ∂A is at $x_1 = 4$, a plane of area $L_y \cdot L_z$.

- Same BP α -integration of §2 with 11-point trapezoidal grid ($\Delta\alpha = 0.1$), $\beta = 2.4$, $N_{\text{therm}} = 500$, $N_{\text{decorr}} = 10$, $N_{\text{samples}} = 80$.

The premise of *méthode B* is that, if a leading area-law signal $\hat{S}_2 = \kappa |\partial A| + b$ exists at this (β, L_x, L_T) , then varying $L_y = L_z$ alone should yield a clean linear fit of slope κ .

A.2 Results and weighted fit

The *méthode B* results are summarised in Table 4. The weighted linear fit (errors in quadrature) gives

$$\hat{S}_2 = \kappa_{\text{fit}} |\partial A| + b_{\text{fit}}, \quad \kappa_{\text{fit}} = -0.404 \pm 0.024, \quad b_{\text{fit}} = +19.08 \pm 0.88. \quad (14)$$

A.3 Why *méthode B* fails: geometric breakdown when $L_y \geq L_x$

The fitted slope $\kappa_{\text{fit}} = -0.40$ is non-physical: a genuine area-law entanglement entropy slope must be positive ($\text{EE} \geq 0$). The mechanism of failure is geometric and is easy to state.

The construction implicitly assumes that the region A is a *slab* in the x_1 direction — that is, that A has depth $L_x/2 = 4$ along x_1 and full extent $L_y \times L_z$ in the transverse plane. For this geometry, the entangling surface ∂A is unambiguously the plane at $x_1 = L_x/2$, of area $L_y \cdot L_z$.

This picture breaks down when the transverse linear extent L_y, L_z exceeds the depth $L_x/2 = 4$ (i.e., $L_y \geq 8$). When the in-plane extent of A is larger than the orthogonal extent, the conventional “area is one face of a brick” picture fails: the deformed temporal connectivity is itself only $T = 8$ thick, while the spatial extent of A is now $8 \times 8 \times 4$ in (y, z, x_1) for $L_y = 8$, or $10 \times 10 \times 4$ for $L_y = 10$. In these regimes, the assumption that $|\partial A| = L_y \cdot L_z$ counts the correct boundary area is no longer valid; the intuitive “area” becomes mixed with edge and corner contributions that are not captured by the simple product.

Empirically, the sign of $\hat{S}_2$ flips between $|\partial A| = 36$ (positive) and $|\partial A| = 64$ (negative) in the *méthode B* table, which is precisely the crossover between $L_y < L_x$ and $L_y \geq L_x$. We interpret this as a clean geometric signal of the breakdown, not as a physical phenomenon.

If we restrict the fit to the two points with $L_y < L_x$ (i.e., $|\partial A| = 16$ and 36), we obtain

$$\kappa_{\text{fit,restricted}} = \frac{S_2(36) - S_2(16)}{36 - 16} = \frac{13.10 - 10.17}{20} \approx +0.147 \pm 0.066, \quad (15)$$

which is at least positive. However, two data points are not sufficient for a meaningful area-law extraction, and the “intercept” $b \neq 0$ (the constant offset includes both the finite-size correction and a possible logarithmic subleading contribution) is then not separately constrainable. A clean *méthode B* in the future would require fixed $L_x \gg L_y$ throughout (e.g., $L_x \geq 16$ with $L_y \in \{4, 6, 8\}$), which approximately doubles the compute relative to the present run.

A.4 Status of *méthode B*

We report *méthode B* as a documented *negative* result. It is not a viable shortcut to area-law extraction at the lattice sizes we have access to. The lesson — that asymmetric lattices need $L_x \gg L_y, L_z$, not $L_x = L_y = L_z$ — is methodologically useful and worth recording.

B V1 vs. V2 sampling comparison: fixed ϵ is biased

B.1 Setup of V1

The V1 production run, performed before the adaptive step (12) was implemented, used a fixed Metropolis step $\epsilon = 0.3$ across all $\alpha \in [0, 1]$. The other Monte Carlo parameters at the time were slightly less aggressive than the V2 ones; in particular N_{therm} ranged from 300 to 600 and

N_{samples} from 30 to 200 (the exact V1 settings are encoded in the V1 source file released with the data; see §8).

B.2 V1 results table

The V1 results are given in Table 5 (only $L \in \{4, 6, 8, 10\}$; $L = 12$ was not in the V1 grid).

B.3 Direct V1 vs. V2 comparison

Table 6 summarises the V1 vs. V2 $\hat{S}_2$ values at the same (β, L) .

B.4 Mechanism: acceptance collapse at large α in V1

The mechanism of the V1 bias is straightforward. At fixed $\epsilon = 0.3$, the per-link Metropolis acceptance rate falls monotonically with α , because the action gradient steepens as the configuration is pulled towards the period- $2T$ conifold. At $L = 10$ and $\alpha \gtrsim 0.7$, we observed acceptance rates below 5% in V1; this means the Markov chain effectively freezes on a 30-sample-per- α run for many consecutive sweeps. The samples that are then averaged into $\overline{\partial_\alpha S_k}$ at large α are not drawn from the α -conditional Boltzmann distribution: they are drawn from a configuration that was last accepted at smaller α and that no longer represents the equilibrium at the current α . The integrand at large α in V1 is therefore *positively biased* (the chain stays in a low-action, positive-integrand basin), pulling the α -integral upward.

This is consistent with the V1 per- α trace showing near-constant values at $\alpha \geq 0.65$ ($L = 10$: $\overline{\partial_\alpha S_k} \approx +30$ to $+100$ across the top half of the α grid, with very little variation, indicative of a frozen chain).

After the V2 adaptive step is in place, the acceptance rate stays $\gtrsim 30\%$ across the full α range, the chain mixes, and the integrand becomes monotonically decreasing through zero around $\alpha \approx 0.55$ (see Table 3), giving the small/negative $\hat{S}_2$ values of Table 2.

B.5 Status of V1

We treat V1 as a documented *biased* run and treat V2 as the methodologically correct one. The V1 data is preserved in the public data archive (§8) for reproducibility and historical reference. We do *not* cite the V1 $L = 6$ value as a measurement of $\log(3)/(2\pi\sqrt{2})$; see §5.

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α	$L = 4$	$L = 6$	$L = 8$	$L = 10$	$L = 12$
0.00	+32.88	+37.07	+64.75	+70.85	+10.02
0.05	+42.24	+20.13	+15.07	+72.64	+45.74
0.10	-0.02	+21.95	+71.72	+31.93	+57.12
0.15	+15.37	+23.76	+94.57	+49.69	+28.50
0.20	+38.10	+18.65	+80.83	+55.87	+77.65
0.25	+34.44	+10.20	+81.66	+63.15	+65.20
0.30	+27.71	+28.60	+33.00	+32.30	+61.68
0.35	+7.34	+28.89	+43.01	+39.63	+11.08
0.40	+28.49	+44.46	+36.27	+65.07	-72.77
0.45	+14.97	+43.22	+39.45	+76.67	-74.89
0.50	+17.19	+32.52	+12.96	+48.58	+12.26
0.55	-9.52	-3.61	-9.13	-28.78	-7.64
0.60	-24.90	-29.93	-25.58	-48.99	-68.91
0.65	-26.98	-24.88	-50.81	-70.34	-107.45
0.70	-21.54	-14.78	-64.28	-55.32	-40.23
0.75	-33.54	-34.13	-71.51	-50.72	-32.07
0.80	-38.96	-74.87	-72.89	-32.95	-52.77
0.85	-46.14	-71.98	-79.90	-49.53	-89.54
0.90	-47.70	-72.54	-107.10	-78.73	-99.55
0.95	-49.58	-110.94	-124.21	-123.12	-176.99
1.00	-61.85	-122.41	-142.67	-180.81	-217.17

Table 3: V2 per- α mean integrand $\overline{\partial_\alpha S_k}$ for all $L \in \{4, \dots, 12\}$ at $\beta = 2.4$. The crossover from positive to negative near $\alpha \approx 0.55$ is qualitatively consistent across $L = 4, 6, 8, 10$; the $L = 12$ profile is noisier with anomalously negative points at $\alpha = 0.40, 0.45$ that we attribute to the small sample count.

$L_y = L_z$	$ \partial A $	$\widehat{S}_2$	$\widehat{S}_2/ \partial A $	geometry valid?
4	16	$+10.17 \pm 0.68$	$+0.636 \pm 0.042$	yes ($L_y < L_x$)
6	36	$+13.10 \pm 1.14$	$+0.364 \pm 0.032$	yes ($L_y < L_x$)
8	64	-3.75 ± 1.79	-0.059 ± 0.028	marginal ($L_y = L_x$)
10	100	-32.04 ± 2.27	-0.320 ± 0.023	no ($L_y > L_x$)

Table 4: Méthode B asymmetric-lattice results at $\beta = 2.4$, $L_x = L_\tau = 8$. The fitted slope is $\kappa_{\text{fit}} = -0.40 \pm 0.02$, which is non-physical (a genuine entanglement entropy area-law slope must be positive). The non-monotonicity in $|\partial A|$ and the sign change between $|\partial A| = 36$ and $|\partial A| = 64$ are diagnostic; see §A.3.

L	$ \partial A $	$\widehat{S}_2$ (V1)	$\widehat{S}_2/ \partial A $ (V1)	Comment
4	16	$+19.24 \pm 0.14$	$+1.202 \pm 0.009$	UV/finite-size dominated
6	36	$+4.39 \pm 0.42$	$+0.122 \pm 0.012$	within 1.3% of $\log(3)/(2\pi\sqrt{2})$
8	64	$+2.26 \pm 0.54$	$+0.0353 \pm 0.0085$	factor ~ 1.5 from Rabenstein
10	100	$+67.18 \pm 1.23$	$+0.672 \pm 0.012$	outlier (acceptance collapse)

Table 5: V1 production-run results, with fixed Metropolis step $\epsilon = 0.3$. *All values are positive*, in contrast with the V2 results of Table 2 where all values are negative. The $L = 6$ value at 0.122 apparently matches the Donnelly–Wall coefficient $\log(3)/(2\pi\sqrt{2}) \approx 0.124$; §5 explains why we now read this as an artefact. The $L = 10$ outlier was the original “sampling” signature that triggered the V2 adaptive-step development.

L	$\widehat{S}_2$ (V1, fixed ϵ)	$\widehat{S}_2$ (V2, adaptive ϵ)	sign flip?
4	$+19.24 \pm 0.14$	-4.38 ± 0.46	yes
6	$+4.39 \pm 0.42$	-10.40 ± 0.77	yes
8	$+2.26 \pm 0.54$	-6.79 ± 1.11	yes
10	$+67.18 \pm 1.23$	-2.90 ± 1.90	yes
12	$— \pm —$	-28.36 ± 3.56	—

Table 6: Direct V1 vs. V2 comparison of $\widehat{S}_2$ at $\beta = 2.4$. Every common L value has its sign flipped between V1 (fixed ϵ , biased sampling at large α) and V2 (adaptive ϵ , unbiased sampling). This is the methodological evidence that V1 was systematically biased and that the V2 protocol is the correct one.